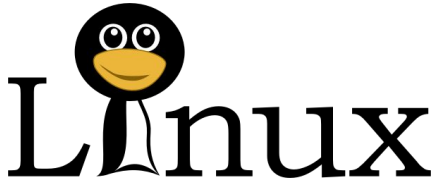
CSP-203: Software Tools & Technologies Laboratory



Lab Session - 4

Piping, Network Command and C Program Implementation of Linux Commands

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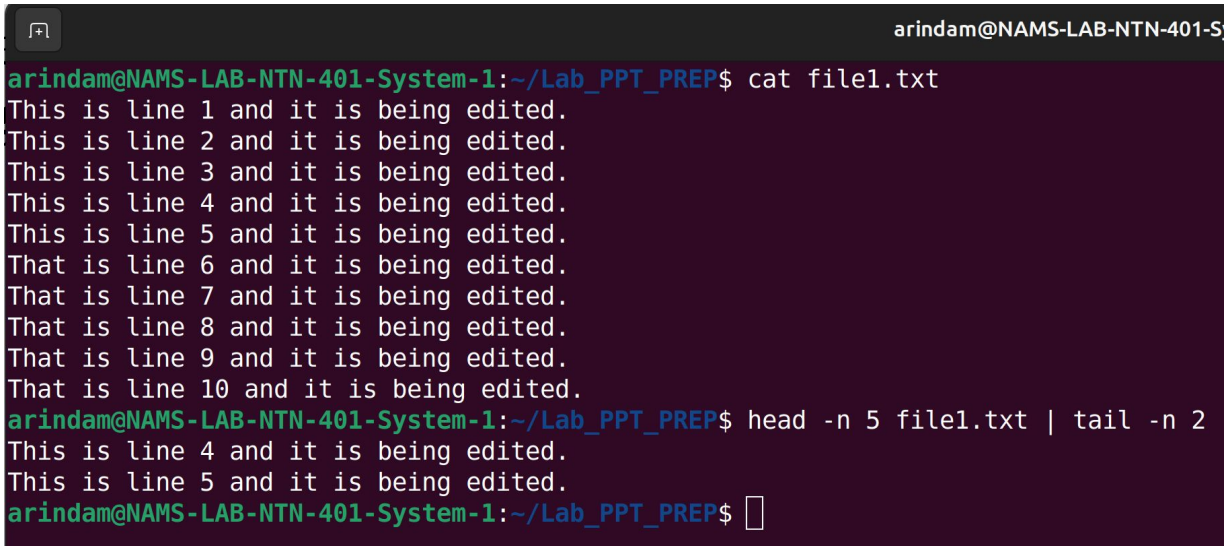
Objectives:

- Piping
- Network Command
- Implementing Linux Commands using C Program

Piping:

Command Sequence: Command 1 | Command 2 | Command 3 | …

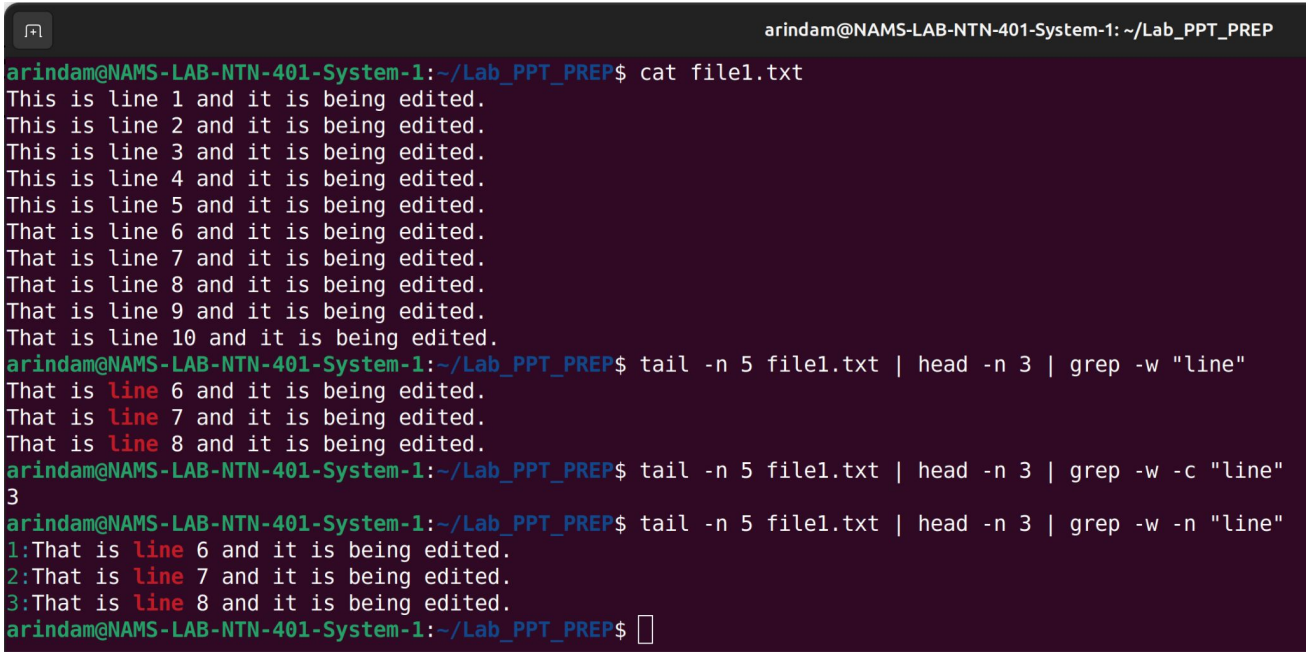
➔ Printing 4th and 5th line with piping: head -n 5 <filename> | tail -n 2



Piping:

Command Sequence: Command 1 | Command 2 | Command 3 | …

➔ Find the occurrence of “line” word b/w 6th to 8th line: tail -n 5 file1.txt | head -n 3 | grep -w -n "line"

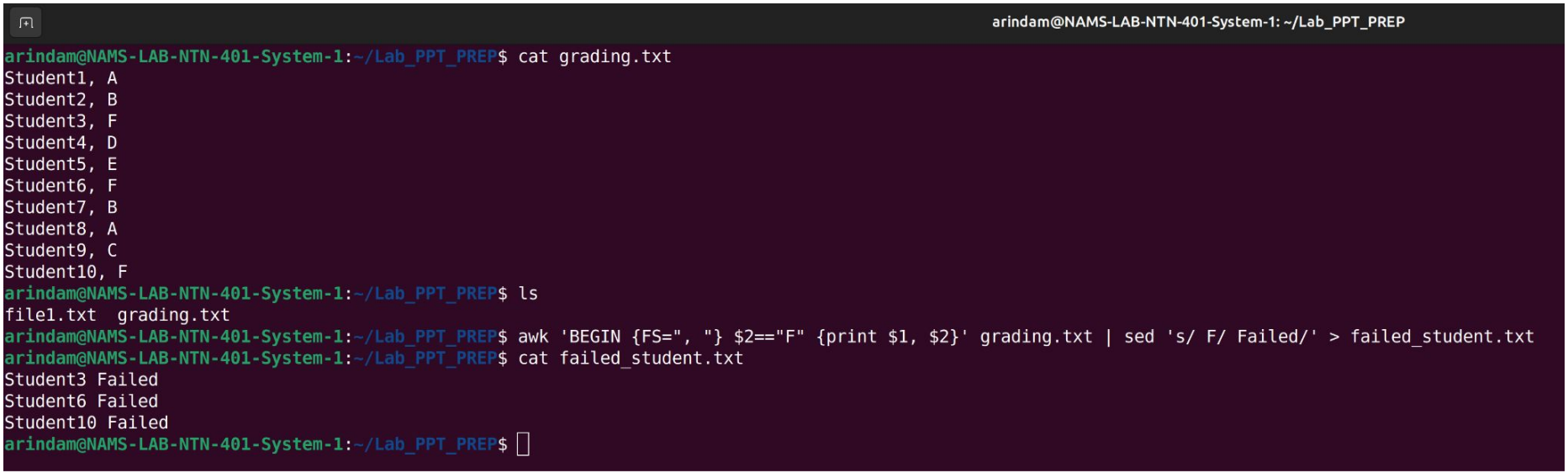


Piping:

Command Sequence: Command 1 | Command 2 | Command 3 | …

➔ Find the failed student list and append the list to another file with replacing F as Failed:

- ◆ awk 'BEGIN {FS=", "} $2=="F" {print $1, $2}' grading.txt | sed 's/ F/ Failed/' > failed_student.txt



Piping:

Command Sequence: Command 1 | Command 2 | Command 3 | …

➔ You want to delete the first student record who has got F:

- sed "/$(awk 'BEGIN {FS=", "} $2=="F" {print NR,$1,$2;exit}' grading.txt | cut -d ' ' -f 1)/'d' grading.txt



Piping:

Command Sequence: Command 1 | Command 2 | Command 3 | …

- ➔ Sorting file list based on size:
  - ◆ ls -l | sort -r -k 5

```
Student9, C
Student10, F
arindam@NAMS-LAB-NTN-401-System-1:~/Lab_PPT_PREP$ ls -l
total 8
-rw-rw-r-- 1 arindam arindam   0 Aug  4 09:32 failed_student.txt
-rw-rw-r-- 1 arindam arindam 391 Aug  4 09:07 file1.txt
-rw-rw-r-- 1 arindam arindam 121 Aug  4 09:18 grading.txt
arindam@NAMS-LAB-NTN-401-System-1:~/Lab_PPT_PREP$ ls -l | sort -r -k 5
-rw-rw-r-- 1 arindam arindam 391 Aug  4 09:07 file1.txt
-rw-rw-r-- 1 arindam arindam 121 Aug  4 09:18 grading.txt
-rw-rw-r-- 1 arindam arindam   0 Aug  4 09:32 failed_student.txt
total 8
```

Networking Command:

ping – Check if a host is reachable and measure latency

ifconfig – View or configure network interfaces (deprecated, use ip)

ip a – Show IP addresses of network interfaces

nslookup – Query DNS to get IP of a domain

traceroute – Show the route packets take to a host

ssh – Secure shell remote login to another machine

wget – Download files from the internet

C program to get output like linux commands:

Your own wc command to print number of lines of a file. Name of your command should be wcl:

Open

```c
#include<stdio.h>

void main(int argc, char *argv[]){
        FILE *fp;
        char c,filename[100];
        int count = 0;
        fp = fopen(argv[1], "r");
        while((c=getc(fp))!=EOF){
                if(c=='\n'){
                        count++;
                }
        }
        printf("%d %s",count,argv[1]);
        fclose(fp);
}
```

```
4 file1.txt
arindam@NAMS-LAB-NTN-401-System-1:~/dir1$ gcc filelinecount.c -o wcl
arindam@NAMS-LAB-NTN-401-System-1:~/dir1$ ./wcl file1.txt
4 file1.txtarindam@NAMS-LAB-NTN-401-System-1:~/dir1$
```

C program to get output like linux commands:

Implement your own wc -w command. Name of your command should be wcw:

Open

```c
#include<stdio.h>
#include<string.h>
#include<stdlib.h>

void main(int argc, char *argv[]){
        FILE *fp;
        char c,filename[100];
        int count = 0;
        fp = fopen(argv[1], "r");
        if(argc < 2){
                printf("No enough arguments...!!!");
                exit(1);
        }
        if(fp == NULL){
                printf("No File exists in this name...!!!!");
                exit(1);
        }
        while((c=fgetc(fp))!=EOF){
                if(c==' ' || c == '\n'){
                        count++;
                }
        }
        printf("%d %s\n",count,argv[1]);
        fclose(fp);
}
```

arindam@NAMS-LAB-NTN-401-System-1: ~/dir1

```
arindam@NAMS-LAB-NTN-401-System-1:~/dir1$ gcc filelinecount.c -o wcw
arindam@NAMS-LAB-NTN-401-System-1:~/dir1$ ./wcw file1.txt
27 file1.txt
arindam@NAMS-LAB-NTN-401-System-1:~/dir1$ wc -l file1.txt
8 file1.txt
arindam@NAMS-LAB-NTN-401-System-1:~/dir1$ wc -w file1.txt
27 file1.txt
arindam@NAMS-LAB-NTN-401-System-1:~/dir1$
```

fseek function:

- ➔ fseek(pointer, offset, position)
  - ◆ Positions:
    - SEEK_END : End of a file
    - SEEK_SET: Starting of a file
    - SEEK_CUR: File pointer’s current position.

- ➔ Try to implement other commands yourself…….


THANK
YOU